CHAPTER

19

Recursive Filters

Recursive filters are an efficient way of achieving a long impulse response, without having to perform a long convolution. They execute very rapidly, but have less performance and flexibility than other digital filters. Recursive filters are also called *Infinite Impulse Response* (IIR) filters, since their impulse responses are composed of decaying exponentials. This distinguishes them from digital filters carried out by convolution, called *Finite Impulse Response* (FIR) filters. This chapter is an introduction to how recursive filters operate, and how simple members of the family can be designed. Chapters 20, 26 and 33 present more sophisticated design methods.

The Recursive Method

To start the discussion of recursive filters, imagine that you need to extract information from some signal, $x[\ ]$. Your need is so great that you hire an old mathematics professor to process the data for you. The professor's task is to filter $x[\ ]$ to produce $y[\ ]$, which hopefully contains the information you are interested in. The professor begins his work of calculating each point in $y[\ ]$ according to some algorithm that is locked tightly in his over-developed brain. Part way through the task, a most unfortunate event occurs. The professor begins to babble about analytic singularities and fractional transforms, and other demons from a mathematician's nightmare. It is clear that the professor has lost his mind. You watch with anxiety as the professor, and your algorithm, are taken away by several men in white coats.

You frantically review the professor's notes to find the algorithm he was using. You find that he had completed the calculation of points $y[0]$ through $y[27]$, and was about to start on point $y[28]$. As shown in Fig. 19-1, we will let the variable, n, represent the point that is currently being calculated. This means that $y[n]$ is sample 28 in the output signal, $y[n-1]$ is sample 27, $y[n-2]$ is sample 26, etc. Likewise, $x[n]$ is point 28 in the input signal,